

FHS Knowledge and Reality: Persistence

Reading

Core Readings

Sally Haslanger and Roxanne Marie Kurtz, eds. (2006) *Persistence: Contemporary Readings* (MIT Press). Intro, chapters 3, 5

Haslanger, Sally (2003) 'Persistence through Time' in Michael J. Loux and Dean W. Zimmerman, eds. *The Oxford Handbook of Metaphysics* (OUP).

Further Readings

Kurtz, Roxanne Marie (2006) 'What's the Problem?' in Sally Haslanger and Roxanne Marie Kurtz, eds. (2006) *Persistence: Contemporary Readings* (MIT Press). Chapters 4, 6,

Hawthorne, J. "Three Dimensionalism vs. Four Dimensionalism" in *Contemporary debates in metaphysics*

Magidor, O. "Endurantism vs. Perdurantism?: A Debate Reconsidered" in *Noûs* 2016

Past Year Questions

'Persistence through time as imagined by those who disbelieve in temporal parts [resembles] a medieval tale that saints were able to practice bilocation.' (LEWIS) Discuss. [2021-2022]

'Events take up time; objects persist through it.' How, if at all, should we explain this difference? [2021-2022]

Are you a sum of temporal parts, an instantaneous temporal part of such a sum, or neither? [2020-2021]

Is there a best account of persistence over time? If so, what is it? [2019-2020]

Part I: The Puzzle of Persistence

Kurtz's take on the Puzzle

Initial tension concerning persistence

- We have some firmly held intuitions about how things are in the world
 - Consistency: Nothing can have incompatible properties
 - Change: Change involves incompatible properties
 - Persistence: Objects persist through change i.e. an object at t1 and an object at t2 is the same object
- The initial tension: holding any two involves the denial of the third
 - Persistence + Change denies Consistency since nothing can have the incompatible properties of being bent and straight
 - We cannot let go of Consistency because it is our strongest held intuition
 - Persistence + Consistency gives up Change → a change nihilist → high metaphysical cost
 - Change + Consistency gives up Persistence
- The fundamental problem is that we want to take consistency, change and persistence to be nonnegotiable metaphysical theses → we need to create an account of persistence that genuinely resolves the initial tension by reconciling the three theses
 - The challenge then becomes: What is it for ordinary objects to persist through change? Is it for them to *perdure*, *exdure*, or *endure*
 - Each approach explains persistence, change, and the involvement of incompatible properties in change in a way that makes them consistent
- In turn, we have the following metaphysical properties
 - Numerical identity: the usual identity relation i.e. the relation every object bears to itself in virtue of it being a single object
 - Survival: an object survives when it is more than a momentary object; more precisely, an object survives iff it is numerically identical to something that exists at a different time
 - Alternation: an object alters by gaining or losing properties; more precisely, an object alters iff it is numerically identical to objects that instantiate different properties at different times
 - Just having a property: an object just has a property if and only if it simply instantiates (Fx) that property. That is, an object just has a property if and only if no extrinsic facts are relevant to the truth of the proposition that the object has that property
- Two broad views
 - Metaphysics of Temporal Parts (MTP): Objects have temporal parts → perdurantism and exdurantism depend on how well they use temporal parts to explain the persistence of objects through change

- Metaphysics of Enduring Things (MET): to claim that some objects endure is to claim that in some cases a numerically identical object is wholly present at different times
 - Also accept that they have temporal parts: MET merely allows for the existence of temporal stages while MTP entails their existence; only MET entails the existence of some enduring objects
- MTP requires objects partially or wholly constituted by stages, while MET requires the existence of objects falling outside that domain
 - For, an enduring object cannot be a space-time worm, nor can it be identical to a single momentary stage; otherwise, it could not be wholly present at different times → so we see that on a view consistent with MET, multiple momentary duck stages could exist but an enduring duck would be distinct from a duck that is either a space-time worm or a single momentary stage
 - A MET theorist may hold that an object identical to a stage or having stages as parts exists only as an extraordinary object or does not exist at all
- Kurtz's 'Real problem of Persistence': we must deny, substantially revise or significantly reinterpret our ordinary understanding of these negotiable theses (Change, Consistency, Persistence) in seeking any solution to the real problem of persistence which is undergirded by the following concepts
 - Alteration (constrains how things change): any object that changes is the proper subject of the incompatible properties involved in the change
 - It says that an object undergoes change only if the persisting thing is the bearer of incompatible properties
 - Survival: If an object persists through change, then the object existing before the change is numerically identical to the one existing after the change
 - Survival constraints the class of objects to those that survive through change: entails that the book persists when it is shut only if is (numerically) the very same book before and after
 - The exdurantist clearly has to negotiate the link between survival and persistent (perhaps through an I-relation)
 - Atemporal instantiation: if an object is the proper subject of a property, then (i) the object has that property, and (ii) facts about time and tense are irrelevant to the truth of the proposition that the object has that property
 - A particular metaphysical view about the properties of proper subjects: an object is the proper subject of a property only if the object instantiates the property without temporal qualification (Fx) rather than Fx-at-t

- The perdurantist and exdurantist appeal to temporal parts metaphysics allows them to mark distinct stages as the proper subjects of the incompatible properties (Fx and not-Fy) making AI unproblematic
- Some metaphysics of time
 - Eternalism: all times exist and thus the objects present at all past, present and future times exist
 - Presentism: only the present exists, and thus that only those objects present now exist
 - Only the eternalist can quantify over all times, and objects existing at those times
- Link to persistence
 - MTP Eternalism: any view of persistence incorporating MTP entails eternalism
 - MET Presentism: any view of persistence incorporating MET entails presentism
 - If we accept these claims, then debates about the metaphysics of time bear on the debate of persistence → denying eternalism denies perdurantism
 - Perdurantists are thought to be unable to refer to other temporal parts without future or past times
- But the entailment fails:
 - Perdurantists could hold that a space-time worm exists only at the present moment and somehow subsists at other times
 - An exdurantist could hold that an ordinary object exists now while its counterparts merely subsist at different times because they already have existed, or will exist, or perhaps that they exist now as abstract representations
- Endurantism and presentism
 - Endurantism's endorsement of MET commits it to a presentist account of time only if the strong claim that all forms of mediated property instantiation are unacceptable is true

Haslanger's Take on the Puzzle

Tension concerning persistence

- Principles that we hold firmly about features of alteration and the world
 - Persistence condition: Objects, such as a candle, persist through change
 - Incompatibility condition: The properties involved in a change are incompatible
 - Law of noncontradiction: Nothing can have incompatible properties
 - Identity condition: If an object persists through a change, then the object existing before the change is one and the same object as the one existing after the change

- Proper subject condition: The object undergoing the change is itself the proper subject of the properties involved in the change i.e. the persisting candle is itself the proper subject of the incompatible properties
- Metaphysics of Time and the debate of persistence
 - Presentist: only the present exists and only present objects
 - Irreducibility of tensed to untensed predication
 - Non-presentist: denies that only the present exists
- Compatibility between different views and Time
 - Presentist endurantism: Only presently existing objects exist; ordinary objects exist by enduring
 - Non-presentist endurantism: Past, present, and future objects all exist; ordinary objects persist by enduring (allowing that there be both enduring, exduring and perduring particulars)
 - Is presentism compatible with exdurantism → since the future counterparts I-related to the present stage do not exist, we may not be able to make cross temporal relations i.e. “I will be 21 next year” → prima facie a problem but Sider gives a way
 - If to perdure an object must exist at different times by having parts at those times, the perdurance is not compatible with presentism → but the presentist could say that a persisting candle consists of the present-candle part and those parts of it that already were and those will be
- Implicit in some understanding of presentism that things existing in the present are wholly present, and the properties they instantiate are only those they presently instantiate
 - If this is true, then the perdurance theorist could not maintain that the candle exists (is wholly present in the present) in a way consistent with both its perduring and presentism, since the candle is a 4D worm
 - So, all perdurance and exdurance theorists prefer to state their positions in a form that favours non-presentism
- Relationalism, incompatibility and temporary intrinsics: assumes eternalism
 - We can change the incompatibility condition – that the properties involved in a change are incompatible – by combining an endurantist account of persistence with a relational approach to temporal qualification i.e. that all properties are really relations to times
 - Being straight at t1 is not a contradiction to being bent at t2
 - Does this sacrifice our understanding of change as possessing incompatible properties?
 - I think it does no worse than the perdurantist
 - Should be more sensitive to what change and the incompatibility condition requires: we can take R and R* to be incompatible just in case nothing can stand in both R and R* to the same things

- So while being straight at t_1 and being bent at t_2 are compatible, being straight and being bent at t_1 are incompatible
- Presentism and just having a property
 - The candle is bent → the candle just has bentness; we want to have this primitive predication of property
 - We can understand the candle was straight as “it was the case that the candle just has straightness” → serious tensing which treats the predicative elements in propositions as tensed rather than tenseless
 - A solution for endurantism that resists relationalism but is also compatible with eternalism
 - But Zimmerman argues that eternalism + serious tensing is not a good combination: suppose that my past headache exists but not presently. Then we cannot say that it is painful since it is not presently painful, and thus we need to say that it isn’t (though it was painful) → but then we’re left with a ghastly non-painful headache
 - Response: obviously, the headache ceased to exist once there was no more pain → it is 1-1 with the pain one feels → eternalism does not mean that the headache lasts forever; it just means the past moment in which it was painful exists

Part II: 3D Views – Endurantism

Endurantism

Summary from Kurtz

- On the straightforward endurantist view, ordinary objects fall within the domain of enduring things of the MET theorist – if a duck is an ordinary object, then it would not be a space-time worm or a single momentary stage
 - Would be wholly present at different moments
 - Multiple momentary duck stages could exist but an endurantist would hold that an ordinary, enduring duck would be distinct from them
- Ordinary objects persist by enduring: identity over time is the strict identity between objects wholly present at different times and change is taken to be the instantiation of incompatible properties by objects identical over time
 - Most intuitive understanding of change over time
- Wholly present ordinary objects have incompatible properties in some sense: but they avoid contradiction by holding that temporal facts mediate the instantiation of incompatible properties without appeal to temporal parts
 - Basically, Book is open at t_0 and shut at $t_1 \rightarrow$ but it generates tension with the view that objects just have the incompatible properties involved in change
 - But finding out what time it is certainly seems irrelevant to determining whether the book is open or shut
- Temporary Intrinsics
 - Intrinsic property of an object is one that the object has simply by virtue of being itself $\rightarrow$ temporary intrinsics are intrinsic properties that an object has only temporarily
 - Apparently, each of these properties – duck's capacity for flight, acorn's colour – is bonded to the relevant object in a way that has nothing to do with anything outside the object $\rightarrow$ properties are in no way mediated or qualified
 - Bradley Regress occurs if we do not have temporary intrinsics:
 - Some objects stand in a relation to each other (1)
 - A relation stands in a relation to the objects that instantiate it (2)
 - Any relation will generate an infinite sequence of relations obtaining among relations, other relations, and objects $\rightarrow$ duck stands in a relation of nearness to the acorn and moreover the relation of nearness stands in the relation to the duck and acorn, as the pair of objects instantiating nearness. And the relation of instantiation stands in a relation of occurring to the duck-acorn pair and the nearness relation, and so on
 - We can accept AI with modification: instead of mediated instantiating reducing to relational forms of instantiation, to bypass the Bradley regress we can argue the following:

- Insist that a relation involving time is not an everyday relation but is somehow special in a way that blocks the regress
- Tensed predicate relations: the book was open, making time-dependent properties (the book is open-at-t)
- Bypasses the Bradley regress but they do not allow us to say that the book just has the temporary intrinsic property of being open because the fact of the book being open turns on some fact about time or tense

Defence against temporary intrinsics objection by Haslanger

- Main objection of temporary intrinsics: Objects are not able to have temporary intrinsic properties which they have in virtue of itself alone, and not by virtue of its relations to other things → properties that any duplicate would have
 - Lewis sees no problem with there being some relational changes such as “when my son grows to be taller than I am”, which is relativised by time
 - The main problem is that we account for all change by construing properties as relational: eliminates temporary monadic properties i.e. temporary qualities, or temporary intrinsics, for apparently monadic properties (such as being bent or being straight)
 - The concern for temporary intrinsics arises because we simply do want to accommodate the phenomenon of intrinsic change, more specifically, to have an understanding of what it is to be a subject of properties that makes intrinsic change possible
 - But what does being an “intrinsic duplicate” even mean and whether being monadic is essential to the notion of an intrinsic property is an open question: consider two balls b and b^* → plausibly intrinsic duplicates can exist at different times, so suppose b exists at t and b^* exists at t^* . Suppose b and b^* differ in their relational properties e.g. b is red at t but b^* is not red at t^* . But surely contrary to our original supposition, we should not count the balls as intrinsic duplicates even if they only vary in the relational ways just indicated
 - But if the balls must be alike in the relational respects in order to be intrinsic duplicates, then it is plausible to say that their intrinsic nature is not captured by their monadic properties
 - Conversely, suppose that no temporary properties are monadic, but x and y stand in all same two-placed relations with respect to times → won't we consider them internal duplicates

Haslanger and Temporally Qualified Instantiation

- Endurantists are sometimes uncomfortable with a presentist ontology and what to allow that times other than the present and things other than presently existing

things exist but also uncomfortable with a relationalist account of temporal qualification

- Solution: an object's having a property at a time should be explicated so that it is the object's having of the property that is temporally qualified
- Coupla tensing
 - Objects have properties → having is open to temporal qualification
 - Candle has straightness-at-t → straightness-at-t is the property itself
 - Criticism: treating the copula as a separable relation lands one in the Bradley regress → if a separate copula is needed to bind an object to a property, then what binds an object to the copula itself (and isn't the separate copula just a relation like others). And if we need another copula to work, then we will need an infinite number of copulas
 - Suggest that the copula is not like other properties or relations and is uniquely able to bind (other) properties or relations to their subjects
- Adverbialism
 - Just having a property is what should be temporally modified while sidestepping the Bradley regress → The candle is straight in the morning → The candle is in-the-morningly straight or the candle is in-the-afternoonly bent
- SOFism
 - Like modal discourse: treat times as possible worlds so "the candle was bent" is true if there was some time t before current time t1 such that at t the candle was bent
 - Can have many different quantifiers over time → but how do we refer to a particular time or range of time
- Structure of the complaint: temporary intrinsics are still not had simpliciter → there is a gap or a mediation between an object and its properties
 - Complaint
 - In order to accommodate intrinsic change, what's predicated of the object (and at issue in the change) must be genuine monadic property (and not a disguised relation)
 - P is a genuine monadic property only if something can have P simpliciter
 - On the various adverbialist accounts the objects don't have the properties simpliciter
 - So, the various adverbialists account cannot accommodate temporary intrinsics
 - Against the SOFist, requires the charge of unqualified predication condition: To say that an object has a property (that x is F) is to say that there is an unqualified predication holding between the object and the property; (and

similarly, what makes it the case that objects stand in a relation (that Rxy) is an unqualified predication holding between the objects and the relation)

Part III: 4D Views – Perdurantism and Exdurantism

Perdurantism

Summary from Kutz

- Perdurantism: the perdurantist takes change over time to be analogous to change over space. Just as colour changes across the surface of canvas when different spatial parts of the canvas have incompatible colours, so the colour of an acorn changes over time from green to brown when distinct temporal parts (stages) of the acorn are green and brown
 - Takes advantage of MTP: ordinary objects are space-time worms composed of distinct momentary stages → just as a candle is extended vertically in space, it is also extended temporally into its past and its future
 - Follows that an ordinary object is partially present at any one moment it exists in virtue of the stage existing at that moment
- How Perdurantism solves the tension: Perdurantist hold that because the book survives change, the book persists by perduring
 - It changes because it has as parts two distinct stages that just have the incompatible properties: one stage is shut, one stage is open (Change)
 - It survives its change in virtue of being numerically identical to the space-time worm extended through time (Persistence)
 - Because no single thing is open and shut (rather, distinct parts are), perdurantists can easily accept Consistency
- Cost of Perdurantism: it gives up our everyday understanding of change as alternation – as a space-time worm, the book in its entirety does not gain or lose a property → instead, objects in a succession (the stages) have incompatible properties in virtue of which the book has changed
 - Strictly speaking, perdurantists deny that we can predicate the properties of being open and shut simply of the book as a whole and so they deny that the book is numerically identical to both an object that instantiates being open and a distinct object that instantiates being shut
 - Change consists in the generation and destruction of momentary stages that are parts of ordinary objects
- Tension between our everyday view that objects just have incompatible properties involved in change and perdurantism
 - Distinct stages that are proper parts of a perduring object simply instantiate incompatible properties (Fx and not-Fx) → but the entire object itself does not instantiate those properties → whatever sense it has incomparable properties, it does so through its relationships to its constituent stages
 - Our intuitions about just having a property mesh more consistently with an approach on which the book as a whole, not some part of it, is open or shut

Lewis's View

- Something perdures iff it persists by having different temporal parts, or stages, at different times, though no one part of it is wholly present at more than one time; whereas it endures iff it persists by being wholly present at more than one time
- Perdurance corresponds to the way a road persists through space: part of it is here and part of it is there and no part is wholly present at two different places
 - Endurantists have no concept of a temporal part → analogy to saying that the whole of the long road is in their little village, for not one single lane of it is missing
- Endurance through time is analogous to the alleged trans-world identity of common parts of overlapping worlds; perdurance through time is analogous to the "trans-world identity"
 - A trans-world individual composed of distinct parts in non-overlapping worlds → perdurance is closer to the Lewisian counterpart theory for modal case
- Decisive objection against endurance: account of the persistence of ordinary things such as people or puddles, is the problem of temporary intrinsics
 - Persisting things can change their intrinsic properties → for instance: when I sit, I have a bent shape; when I stand, I have a straightened shape → both shapes are temporary intrinsic properties → I have them only some of the time
 - How is such change possible? I know of only three solutions
- Solution 1: shapes are not genuine intrinsic properties → disguised relations which enduring things bear to times
 - One and the same enduring thing may bear the bent-shape relation to some times, and the straight-shape relation to others
 - In itself, considered apart from its relation to other things, it has no shape at all
 - Lewis thinks that this is an incredible result, if we are speaking of the persistence of ordinary things since we think that shape is a property, not a relation
- Solution 2: the only intrinsic properties of a thing are those it has at the present moment → other times are like false stories/abstract representations composed out of the materials of the present
 - When something has different intrinsic properties according to one of these ersatz other times, that does not mean that it, or any part of it, or anything else, just has them → no more so when a man is crooked according to the Times
 - This solution rejects endurance, because it rejects persistence altogether → Lewis thinks this is worse (i.e. rejecting endurance)
 - In saying that there are no other times, as opposed to false representations thereof, it goes against what we all believe

- Solution 3: the different shapes, and the different temporary intrinsics generally, belong to different things
 - Perdurantism (which Lewis accepts) → we are made up of different temporal parts and our temporary intrinsics are properties of these wherein they differ one from another
 - No problem at all about how different things can differ in their intrinsic properties

Haslanger's View on Perdurantism and Proper Subjects

- Proper subject objection: the object undergoing change is itself the proper subject of the properties involved in the change
 - Perdurantism seems to sacrifice the proper subject condition since it is temporal parts that are the proper subject of the properties involved in change
 - Implies that being bent and being straight are intrinsic properties of candle-stages, railroad-track-stages etc
- No change objection
 - Version 1: the perdurantist view is committed to a static conception of time: time and all its occupants are stretched out 'timelessly' in four dimensions → nothing even comes into or goes out of existence
 - Version 2 (no alteration): stages are thought to come into and go out of existence instantaneously in succession → but there is nothing that gains or loses one of its properties, so there is no genuine alteration
 - Temporal parts come and go, but do not alter
 - But this can arise for any externalist account for the endurantist account or even the endurantist
 - The perdurantist can respond by arguing that change occurs when an object's temporal parts have incompatible properties and can also say that alteration can be understood as variation in the intrinsic qualities of a thing → having a straight-stage and a bent-stage is our best account of such variation
 - No change account does not force us to abandon the eternalist account of change

Endurantism

Summary from Kurtz

- Endurantist or stage theorists, takes identity over time to be analogous to identity between possible worlds
 - Endurantists assume that a duck that now can fly was land-bound in virtue of a flightless counterpart waddling around in the past vs Humphrey could've won the election

- In both cases, distinct objects have incompatible properties
- Exdurantist contends that change over time is nothing more than an object and its temporal counterpart having incompatible properties and existing at different moments
- Exdurantists differ from Perdurantists in denying that ordinary objects are composed of multiple stages and are only partially present: they hold that an ordinary object is numerically identical only to a single stage and that a temporal counterpart is numerically identical to a distinct stage → each stage is wholly present at the moment it exists and exists only at that moment
 - Objects persist by exduring and exdure by changing over time; an object changes over time, when it and a counterpart stage just have incompatible properties
 - It changes over time in virtue of standing in a counterpart relation to a stage from a different time
 - And, because no single thing is open and is shut, exdurantists can easily accept Consistency
- Exdurantism rules out alteration: distinct book stages are open and shut, no one book has lost one property and gained the other
 - Excludes the survival of persisting objects – no book is numerically identical to both the earlier open stage and the later shut stage
- Derivative forms of existence: exdurantists could say “the shut book existed in the past as an open book” is true in virtue of an open counterpart book existing at a different time
 - But this threatens the coherency of the very idea of existence → posits the existence of ordinary objects at times in the world during which they could not have causal powers and could not overlap with any material object

Sider's Stage Theory

- Motivated from the Lewisian solution to Parfit's problem of fission: On his version of the worm view, the I-relation holds between person stages iff they are parts of some one continuing person and that the two people were overlapping people that split up after the fission case → if Ted divides into Fred and Ed, then Ted was just ambiguously referring to Fred and Ed before when they overlapped
 - Lewis claims that the I-relation and the relation of psychological continuity are one and the same relation but Parfit argues that the I-relation does not capture commonsense platitude that identity is what matters
 - Sider thinks one possible understanding of the commonsense platitude is: A person stage matters to my present stage iff it bears the I-relation to my present stage (I1)
 - Unless persons are person stages, which they are not for Lewis, then it does not really make sense to endorse (I1)

- Doctrine that psychological continuity is what matters:
 - For any person P and any person P* existing at some time in the future, $M(P^*, P)$ iff P's current stage is psychologically continuous with P*'s stage at that time
- Doctrine that identity is what matters:
 - For any person P and for any person P* existing at some time in the future, $M(P^*, P)$ iff $P=P^*$ (I2)
 - For any person P and for any person P* existing at some time in the future, $M(P^*, P)$ iff P's current stage bears the I-relation to P*'s stage at that time
- Parfit notes that combined with (PC), (I2) rules out the possibility of a stage of one person being psychologically continuous with a stage of another person, and is thus inconsistent with Lewis' approach to fission
- (I3) avoids the problem but at the cost of avoiding to capture the spirit of commonsense platitude that what matters is identity: "The shared stage [T3] does the thinking for both of the continuants to which it belongs. Any thought it has must be shared"
 - Sider argues that this implies that what happens to Fred matters to Ed (not what they are capable of desiring): even if I cannot desire to survive because I am comatose, it still matters to me that I am comatose
- According to Sider, Lewis has two main responses
 - Speaker of what matters to stages or saying that Ed is incapable of desiring things uniquely on behalf of himself
 - But the latter is not successful
 - The stage view reconciles both ideas about what matters
- Sider's Stage View:
 - Basically people exist momentarily as person stages (and thus cease to exist immediately), but we can analyse a tensed claim "Ted was a boy" as saying that there is some past person stage, x, such that x is a boy, and x bears the I-relation to Ted
 - So despite being a stage, Ted was a boy
 - Given the stage view's counterpart theory of temporal properties, we can accept that PC is what matters and the following doctrine of identity is what matters
 - For any person P and any person P* existing at some time in the future, $M(P, P^*)$ iff P will be identical to P* then (I4)
 - What happens to a person in the future matters to me iff I will be that person → cannot say that a person must timelessly be me for I am not identical to persons at other times but I will be identical to persons at

other times, for I bear the I-relation to future stages that are identical to such persons

- I4 adequately captures the spirit of commonsense platitude that identity is what matters for it says what matters to me is what will happen to me
- This view implies that what happens to Fred matters to me and what happens to Ed matters to me → I will be Fred and I will be Ed → does not mean I will be both Fred and Ed nor does it imply that Fred is identical to Ed
 - We can instead think of what the stage view entails as: futurely-being-F(me) and futurely-being-G(me)
 - Merely says that I am I-related to some future stage that is identical to Fred, and also that I'm I-related to some possibly different future stage that is identical to Ed
- Counting: the implication of Lewis's view that there are two people occupying the same spatio-temporal region is preposterous
 - Lewis defends an unorthodox view of counting → if roads A and B coincide over a stretch that a person must cross, when she asks how many roads she must cross to reach her destination, we can say "one"
 - Counting by "identity-at-t" relation → but it seems that part of the meaning of counting is by identity → when we count a group of objects, we are interested in how many numerically distinct objects there are
 - For the stage view: before division, when I am alone in my room, there is but one stage and therefore one person, if persons are stages
 - Analogy to counterpart theory: I exist only in the actual world, my other-worldly selves are distinct objects related to me merely by counterpart relation
- Objections
 - Stuff that happened to me really happened to someone else instead
 - Reply: Stage view does not have this consequence – "Ted was once a boy" attributes a certain temporal property of 'once being a boy' to me and not to anyone else
 - The stage view does analyse my having this property as involving the boyhood of another object, but I am with the one with the temporal property
 - I cannot say "I am that irritating young boy" because that comes out false → your current stage is not identical to that past stage → Sider bites the bullet
 - We can only say "I was that irritating young boy" and we mean this anyway
 - Nothing persists through time
 - Sider thinks that if we mean "Ted exists at more than one time", then the stage view does have the consequence of no persistence

- But in another sense of persistence, because the stage view allows for temporal predication, the stage view allows that I both exist now and previously existed in the past
 - I.e. survival reduces to merely being I-related to some past stage → we can justify this by saying that survival is isomorphic up to what matters to us
 - Semantic objection by Perry: “[on the stage view] the little boy stealing apples is strictly speaking not identical with the general before me...[the stage view] denies what is clearly true: that when I say of someone that he will do such and such, I mean that he will do it. The events in my future are events that will happen to me, and not merely events that will happen to someone else of the same name”
 - Sider maintains that the stage view is perfectly consistent with stages having temporal properties: it is just that these properties are analysed in terms of the I-relation and other stages
 - Perry finds the stage view’s analysis of these facts as what is objectionable but has two causes for concern
 - Concern 1: the fact that I was once a child and will one day be a man is a fact about two different objects
 - Concern 2: stage-view’s analysis of temporal properties is flat-out implausible → the property being I-related to some stage in the past that is F is just not the same property as the property previously being F
 - Sider just disagrees with this
 - The stage view also has a counting problem: if we take some interval of time [10am to 12pm] → how many people have been sitting in my office → infinitely many stages between the two times assuming that time is continuous
 - Stage view should be restricted to the claim that typical references to persons are to person stages → but in certain circumstances, such as when we take the timeless perspective, reference is to worms rather than stages
- Stage-theorists treatment of proper names: the referent of that name, relative to any time would be the one and only stage that has that property
 - Given a name, such as “Ted,” I’ll speak of stages with the associated individual concept as being “Ted-stages” → must be emphasised that talk of these individual concepts doesn’t require a descriptivist view of reference → name is instead introduced by an initial baptism, where it is affixed to some stage → in normal cases without fission, that baptism completely determines what the referent of the name will be at any later time → i.e. the stage existing at

that later time (if there is one) that bears the I-relation to the originally baptised stage

Hawley's Exdurantism

- Stage theory says that nothing is wholly present at more than one moment but also that the satisfiers of sortal predicates like 'is a banana' are momentary things
 - Thinks that we ought to give up the central endurance claim that things wholly present at different times can be identical and adopt the four-dimensional metaphysical framework
 - Exdurance avoids problems confronting perdurance theory → make stage theory compatible with a wide range of views about properties and avoid commitments to the existence of universals or tropes
- Link between properties and stage theorists
 - The stage theorists can simply claim that whatever it takes to be a banana, stages have what it takes
 - The brief stages satisfy the sortal predicate "is a banana" under the view that properties are universals or classes or tropes
- Difference between stage theory and perdurance
 - The perdurance must claim that if an object is to satisfy a sortal predicate (or instantiate a sortal property), then that object must not be a proper part of something which falls under the sortal → because its temporal part is the banana, rather than the 4D worm being the banana
 - If a persisting object is to satisfy predicates at times in virtue of the properties of its temporal part, then it must not be a proper part of another such predicate-inheritor → does not apply to stage theory
 - According to stage theory, a stage satisfies a sortal if and only if it has the appropriate intrinsic properties and stands in suitable qualitative relations to other stages
- Stage theory comes close to satisfying the endurance theory intuition that we cannot speak atemporally about the changeable properties and parts of persisting objects
 - According to perdurance theory, we can think and talk atemporally about the changeable properties of persisting things – i.e. the apple is wholly (since permanently) green and that the banana is part green and part yellow
 - Endurance theorists think that we need to index changeable properties to time
 - The stage theorist can speak temporally about perduring objects, as perdurantists say we can, and we can make atemporal claims about parthood → but we cannot speak temporally of 'Tony Blair' for it does not refer to perduring 4D things but different objects when used with respect to different times

- Stage theory comes close to accepting that ordinary objects satisfy the endurantist clause that statements about what parts the object has must be made relative to some time or other
- Question 1 for the stage theorist: can instantaneous stages instantiate the properties and satisfy the predicates we associate with ordinary objects?
 - We could say that nothing could be a banana unless it grew on a banana tree → then the instantaneous stage of a banana cannot be said to satisfy the predicate of being a banana
 - Historical predicates
 - Another type of problem: lingering predicates i.e. “is thinking about Vienna” etc
 - Hawley thinks that stages can satisfy both – while an isolated stage cannot satisfy such predicates, an instantaneous stage can satisfy such predicates if it is suitably surrounded by and related to other stages with appropriate properties
 - Roughly speaking: the stage satisfies ‘is travelling across the court’ iff it is embedded in a series of stages which are appropriately located at points across the court
 - The upshot of this is that these lingering or historical predicates become relational – they are relational properties of stages
 - This view does not mean that only collections of stages satisfy predicates – a stage satisfying the predicate “is a banana” is a little like what it takes for a person to satisfy a predicate “is a sibling” → in a suitable relation with some other stage
 - The question that immediately follows is what are these ‘suitable’ relations between stages → exactly what relation must a stage bear to other banana stages in order to qualify as a banana
 - Sider would say the I-relation → criteria of identity throughout time
 - Circularity: to be a banana it is necessary to be suitably related to other stages which are bananas. But those other stages qualify as bananas only because they themselves are suitably related to bananas and so on
 - Hawley thinks this is not a vicious circle → the type of characterisation she has in mind is: for a stage to be a banana, it is necessary that it be yellow, curved, edible, and suitably related to other yellow, curved and edible stages → only the whole thing is perhaps necessary
 - The banana will have both appropriate non-relational properties, and of being suitably related to other things with appropriate non-relational properties
 - I do not see how this solves the problem since Hawley admits the non-relational properties are not sufficient

- Question 2 for the stage theorist: can we refer to instantaneous stages at all, given their brevity?
 - How can we ever refer to ordinary objects if they are fleeting stages and how can we introduce names or descriptions of objects at one time and then use them to talk about other times?
 - Hawley: Stage Theory does not make special demands on theories of reference and hence on this count, is at least as good as endurance or perdurance theories
 - The core question is how do we achieve precise reference of one stage and not another fleeting stage
 - Dangerous to say we can do so as we talk about present moments → but we may never refer to present moments at all
 - Tense-predication → the utterance of “the banana is now yellow” is made true by the presentness – at the time of utterance – of the banana being yellow; while the tenseless theorists say that the utterance is true if and only if the banana’s being yellow is simultaneous with the time of utterance
 - On either account, present-tense predications single out a time – time of utterance – with respect to which they should be assessed
 - Likewise, past and future tense predications single out times with respect to which they should be assessed
 - E.g. ‘the banana was green’ → tenseless theorists say the utterance is true iff the banana’s being green is earlier than the time of the utterance while the tensed theorist says the utterance is made true by its pastness – at the time of utterance – of the banana being green
 - So the stage theorists says the utterance that “the banana is yellow” is true iff the stage which is the banana at the time of utterance is yellow
 - But there might be no unique moment of utterance
 - Hawley thinks the endurantists and perdurantists face equally tricky questions → exactly which temporal part of the banana must be yellow if the utterance or exactly which moment must the banana be yellow if the utterance is to be true
 - I think it's disingenuous to say that the endurance faces an equally difficult question
 - Three possible responses
 - Interpret a given present-tense predication as an existential claim over a domain of moments, temporal parts or stages
 - Interpret it as a universal claim over a domain of moments, temporal parts, or stages
 - Be a supervaluationist and take a present-tense prediction as concerning a single, but vaguely specified, moment, temporal part of stage within a certain domain

- If the predicate is satisfied in the relevant neighbourhood of stages, then it comes out as super true
 - Count the utterance as true if the predicate in question is satisfied by every member of that range, by every potential referent
 - But there will be vagueness in which moment is the salient period
- Problem of re-identification: how does one persisting banana rather than another gets picked out – what determines that we are talking about this banana here rather than that banana over there → Hawley thinks no particular problems here
 - According to Kripke, a name is attached to a persisting object at some time, either by ostension, or by a reference-fixing description (what Sider calls a baptism)
 - The name is passed on from one user to the next and this ‘causal chain’ somehow ensures that in its future uses, the name refers to the object it was first attached to more or less independently of the states of mind of the subsequent users
 - What about Sider’s use of the I-relation?
 - Stage theorists only claim that the name refers to different stages when it is used to talk about different times, but these stages are suitably related to one another → when I talk about Matthew in the future, it refers to some future stage suitably related to Matthew
 - This is no different from the perdurance theory → no banana is wholly present at the dubbing event → a name gets attached to the persisting banana because later parts of the banana are suitably related to the temporal part which is present at the dubbing event

4 Dimensionalism vs 3 Dimensionalism

Summary from Hawthorne

5 thesis the 4D theorist accepts

- Pointiness: Every space-time region is composed of point-sized parts
- Universality: Every collection of space-time points composes something
- Uniqueness: For every collection of space-time points there is exactly one thing that is composed of the space-time points in that collection
- Pointy Object Fundamentality: Points are ontologically more fundamental than extended regions
- Pointy Fact Fundamentality: Facts about points and the relations between them are more fundamental than facts about extended regions

Objections against 4D (with and without spatiotemporalism)

- Spatio-temporality: we make identity claims that a poached egg is identical to a region of spacetime
 - Objection: violates common sense and modal properties of space-time regions as having their spatiotemporal profile essentially, but of their occupants as having their spatiotemporal profile only accidentally
 - Defensive strategy: modal properties are context-dependent → so when I call the poached egg a poached egg, it makes sense that the poached egg does not have the property of being so-and-so-sized essentially, but when I refer to the poached egg via its spatio-temporal profile
- 4D without spatiotemporalism: Do not need to identify ordinary physical objects with space-time regions → the relation of physical objects to their parts is strongly analogous to the relation between spatiotemporal regions and their parts
 - Physical objects are thought to be built out of simple material parts that have zero spatial and temporal extent, and any collection of such simples is held to make up one and only one material object
 - Objections: where a quantity of steel and a ship come in and out of existence at the same time, it would seem that the quantity of steel and the ship will have to be identified

3D picture and objections

- Plenitude: holds that everything is made of point-sized things, and that every collection of point-sized things makes up something, but allows that there may be a number of things made of the very same point sized things
 - Even when a quantity of steel and a ship made of the steel have exactly the same point sized parts, they are not identical
- Objects as wholly present: we motivate this by calling into question the plausibility of a tenseless language
 - "I was happy five minutes ago" → but some suppose the tenseless view of reality is maximally perspicuous
 - Those who take tense seriously as descriptive of reality will question the idea of my being spread out in time → all the present places exists, but for the serious tensor, the claim that I occupy each of a series of past present and future times is the sort of claim that is supposed to be understood in a tenseless language
 - The serious tensor can accept that I am currently mereologically coincident with an instantaneous object, and that it will always be the case that there exists some instantaneous object or other than I am coincident with
 - A moving point rather than a straight line seems to be a more natural description of the state of the world
- Notion of wholly present

- Way 1: Identifies exactly occupies some spatiotemporal region r with wholly occupies → but it is obviously false that a persisting object wholly occupies any spatiotemporal region corresponding to a single instant of time
- Way 2: Idea of wholly located is connected to intrinsicity → natural to distinguish facts that are intrinsic to a time – facts that have to do with how things are at that time, considered in itself – from what is extrinsic to that time
 - Someone who thinks that I am wholly located at each time that I exist maintains, as it were, that God could recognise me as present at any given time that I exist just by considering the world at that time, as it is in itself
 - Makes heavy reliance on the intrinsic/extrinsic distinction, but also supposes that such a distinction is applicable to questions about individual essences – such as whether a property of containing Nixon is intrinsic or extrinsic to this or that time

Sider's Argument for Four Dimensionalism

Basic Position: Mereological Argument, from unrestricted mereological composition, that every object has a fusion across time i.e. has a diachronic fusion or D-fusion

- Ted Sider argues from unrestricted mereological composition, the position that all potential cases of composition compose a whole, to the unrestricted diachronic fusion of perduring wholes composed of temporal parts.
 - Since it cannot be indeterminate whether a class of objects fuse, and any possible restriction on fusion would make it a vague matter whether fusion occurs, Sider argues that it must be the case that fusion is unrestricted: that fusion always occurs.
 - This conclusion guarantees the existence of temporal parts. Further, persisting objects perdure, that is, they are extended in time like spatial objects are extended in space – by having distinct parts in different locations along a dimension
- Definitions
 - Four-dimensionalism is a thesis concerning the nature of how objects persist through time: it is the view that persisting objects perdure – that they are extended through time and composed of instantaneous temporal parts
 - Sider claims that the four-dimensionalist thesis is that every persisting object, X , has temporal parts X_1 and X_2 at times T_1 and T_2 , the union of which compose the timespan of the object X .
 - Three-dimensionalism is the view that objects persist by enduring - by being “wholly present at each time that they exist”
- In the argument from vagueness, Sider understands four-dimensionalism as a thesis concerning temporal locality: a claim that, necessarily, for any object (suppose Ted),

and any distinct, nonempty sets of times such that their union comprises the time-span of the object (suppose T1 and T2), there will be two objects, (Ted1 and Ted2) such that Ted1 has all the parts of Ted at T1, and Ted2 overlaps Ted at T2

- By definition, the objects Ted1 and Ted2 are instantaneous temporal parts.
- The temporal locality thesis thus claims that all persisting objects necessarily have instantaneous temporal parts, and are extended in time by possessing them
- While three-dimensionalists tend to index parts to time, the four-dimensionalist understands parthood atemporally, seeing temporal parts as parts of wholes simpliciter. To bridge the gap in terminology, Sider proposes a purportedly neutral definition of temporary parthood as: Necessarily, X is a part of Y at t iff: (i) X and Y each exist at t, and (ii) X's temporal part at t is part of Y's temporal part at t
- Roughly, the case for unrestricted composition runs as follows: If it is not the case that every class of objects has a fusion, then we can conceive of two possible cases, one in which composition occurs and one in which composition does not occur, which are connected by a continuous series of extremely similar cases. Since composition can never be vague, there must be a clear point in this series where composition abruptly ceases to occur. Such a cut-off is either vague, arbitrary, or anthropocentric, so composition must always occur. Sider's argument is as follows:
 - P1: If not every class has a fusion, then there must be a pair of cases connected by a continuous series such that in one, composition occurs, but in the other, composition does not occur.
 - P2: In no continuous series is there a sharp cut-off in whether composition occurs.
 - P3: In any case of composition, either composition definitely occurs, or composition definitely does not occur
 - C: Every class has a fusion.
- The minimal D-fusion argument exactly parallels the argument for unrestricted composition, substituting assignment (takes one or more times as arguments and assigns nonempty classes of objects that exist at those times as values) for case, and minimal D-fusion for composition
 - Reaches the following conclusion: Every assignment has a minimal D-Fusion
- Sider's motivation for Premises
 - P1:
 - P2:
 - P3: In defense of P3, Sider launches a sub-argument which hinges on two presuppositions – first, that vagueness is a product of semantic indecision rather than *genuine ontological vagueness or ignorance*, (call this condition V) and second, that logic can never be a source of vagueness (call this condition L)

- Criticisms

- Sider accepts that P1 is vulnerable to a nihilist stance about composition: that there are no 'composite' objects, so we cannot say that there is a composition in the first place. In other words, when we speak of tables and chairs, we are saying something more complicated about a collection of particles that 'compose' a 'table'.
 - Reply: Sider rejects Inwagen's nihilism by suggesting that Inwagen relies on the assumption that objects are composed of primitive atoms without proper parts → but Sider thinks that we can always have smaller and smaller objects that are proper parts of what we think are elementary particles without proper parts
 - But Sider's argument against Inwagen seems to rest on what he suggests is an "empirical possibility" that electrons, quarks which are composed of smaller parts, which in turn could composite. It seems to me that the reason why this is compelling against synchronic fusion is that there has been inductive evidence that we repeatedly find that what we assume to be elementary particles are in fact composite. The balance of probability and expectation is thus in Sider's favour
 - But this does not hold for temporal parts – the balance of probability is not in Sider's favour because it is (perhaps impossible) unlikely that we'll ever discover temporal parts empirically. While Inwagen's argument may not hold for composition in the 3D space, we can see that it becomes stronger when applying it to time
- Superfluous objection: (i) An object may persist by enduring or perduring. , (ii) If an object persists, it will have temporal parts, (iii) These temporal parts compose a perduring object which is at best spatially coincidental with, but distinct from the object itself, which may persist by enduring, (C) Not all objects must persist by perduring.
 - This objection maintains that granting unrestricted composition and unrestricted diachronic fusion is entirely compatible with three-dimensionalism. While this much is not to claim the truth of three-dimensionalism, it does undermine Sider's argument from vagueness as a necessary truth.
 - Suppose there is a case of unrestricted diachronic fusion. Let Ted refer to one of the many synchronic fusions guaranteed by unrestricted composition (that is, objects not assigned to times). Suppose now that Ted exists at T1 and T2. Sider's argument for unrestricted minimal D-fusion guarantees the existence of some minimal diachronic fusion Ted1 such that it has the value Ted at T1.
 - Unrestricted composition means that there is an object composed of Ted1 and Ted2. Call this object, Ted's life. The existence of this object,

which perdures and is extended temporally, is entirely compatible with the notion that the initial object, Ted, otherwise persists by enduring. Ted's life, is merely spatially coincidental with Ted. Thus, Sider's argument only guarantees the existence of an event-object, which is at best the history of a thing-object.

- To hold this, the three-dimensionalist will need to explain material coincidence, but this is a plausible auxiliary commitment

Magidor: that the debate between endurantism and presentism rests on strictly orthogonal assumptions/doctrines

Definitions

- Liberalism: the view that for any set of space-time points occupied, there is an object which occupies all the points in that set and no other points
 - Restrictiveness is the denial of liberalism
 - Liberalists must accept that there is an object which is located where my desk is but exists only for an instant; that there is an object located exactly where my desk and my nose are, but nowhere else and so forth
 - Traditionally, Perdurantists are Liberalists and Endurantists are Restrictivists
- Restrictive Perdurantism: Perdurantism requires that there is an object which is at t that is just like Desk, but exists only for an instant; but on its own, the doctrine does not require any of the following – the existence of a point-sized object located in the corner of my desk, the existence of an object which is just like my desk between three and four o'clock but exists only at those times
- Liberal Endurantism: although liberalism does entail the existence of instantaneous objects such as Desk-at-an-instant, it does not entail that it is part of Desk (at the relevant time)
 - LE admits that for any object x and any time t at which it exists, there is an object y that is co-located with x at t and exists only at t, but it typically denies that x ever has any instantaneous temporal parts
 - Endurantism do not deny (i) the possibility of instantaneous objects nor (ii) that distinct objects can have exactly the same spatial location at the same time

1. Argument from Anthropocentrism

- **Objection:**
 - Perdurantism is seen as less anthropocentric because it treats temporal and spatial dimensions equivalently (objects have temporal parts just as they have spatial parts).
- **Endurantist Revision:**
 - **Liberal Endurantism:** Accept a broader ontology of objects (e.g., admitting various unconventional entities that exist wholly at multiple times).

- This revised endurantism becomes equally objective and non-anthropocentric, effectively matching the perdurantist's neutral stance.

2. Argument from Vagueness

● Objection:

- Sider maintains that facts about composition entail facts about how many objects there are, so vagueness about composition entails vagueness about how many objects there are (count-indeterminacy)
 - But since one can formulate claims about how many objects there are using purely precise logical vocabulary, this conclusion is untenable
 - Upshot: in order to block the argument from vagueness, we must take an extreme position to the questions of composition
- Sider's diachronic universalism entails perdurantism only if we assume the Principle of Strong Supplementation: the principle of diachronic universalism entails that for any object O and time t at which O exists, there is an object z which exists at t and only at t, such that z is the fusion of {O} at t
 - It remains to show that z is an instantaneous temporal part of O at t, and in particular that z is a part of O at t which follows if we accept SSP

● Endurantist Revision:

- We can construct an SSP-Free LE that rejects SSP i.e. that short Desk and Desk at t can be composed of the same particles, but Short Desk is not a part of Desk
- Infinite domains: only counting vagueness is there are finite objects i.e. altogether n or n+1 objects → but on the perdurantist account, there would be usually infinite many objects and distinct ITPs
 - Restricting it to a rare scenario where no objects persist for any interval of time does not look helpful
- Even if we accept the finiteness assumption, the assumption from vagueness about composition makes for count-indeterminacy fails against a background of a sufficiently liberal setting → start composition in a purely spatial setting and suppose that there are exactly three spatial points: a liberalist would maintain that for any non-empty subset of these points, there is an object exactly located at those points. Assuming that there is only one and such object located at each spatial region, there will be seven objects all together
 - but even keeping fixed these facts about the number of objects and their locations, there are many different possibilities concerning the parthood and fusion relations amongst these objects → maintain that all seven objects are simples or that objects have an isomorphic mereological-structure to that of the spatial regions they occupy
 - Generalise to time → facts about composition do not make a difference to 'count-facts'

3. Argument from Recombination

- **Objection:**

- Perdurantism allows easier recombination of parts over time, explaining how objects persist despite changing parts.
- When objects **change parts over time** (for example, replacing one of a ship's planks each day), perdurantism easily explains persistence: the object simply has different temporal parts at each moment. Each temporal part has its own configuration of physical parts
 - Perdurantism makes this recombination easy because each slice can differ independently. An object persists by having a continuous "chain" or "series" of temporal parts, even if the actual physical parts differ significantly from one moment to another.

- **Endurantist Revision:**

- Liberal endurantism similarly accommodates recombination by acknowledging flexible criteria for persistence and accepting multiple entities corresponding to different stages.
- Objects persist through part changes without temporal parts by broadly defining persistence criteria.

4. Argument from Temporary Intrinsic

- **Objection:**

- Perdurantism neatly explains how an object can have contradictory intrinsic properties at different times by assigning these properties to separate temporal parts.

- **Endurantist Revision:**

- Endurantism can respond by using time-indexed properties (an object has "property-at-time-t").
- Alternatively, endurantists might simply accept a property relativization strategy, maintaining consistency without temporal parts.

5. Argument from Coincidence (Temporary Coincidence)

- **Objection:**

- Perdurantism effectively distinguishes coincident objects (e.g., statue and lump of clay) by assigning different temporal parts to different objects, even if they coincide spatially.

- **Endurantist Revision:**

- Endurantism handles coincidence by allowing two objects to share spatial parts entirely but differ via non-spatial, non-temporal properties (such as modal or sortal properties).
- This approach makes coincidence manageable without temporal parts.

6. Argument from Motion (Pro-Endurantism)

- **Objection (Endurantist advantage):**

- Endurantism intuitively captures objects as wholly moving through space over time.
- Perdurantism seem to entail numerous violations of the laws of physics: two objects O1 and O2 that are extremely far apart and consider an object which is a fusion of O1's ITP at t, and O2's ITP at an arbitrarily short time after t → given right values, the fused object would have to travel faster than the speed of light
- Also faces problems accounting for the Kripke-Armstrong rotating disc problem
 - It seems possible for there to be a perfectly homogenous disc which is stationary and possible for there to be a duplicate homogenous disc which is rotating
 - Standard perdurantism has trouble distinguishing these possibilities → the disc counts as stationary in how their spatial parts move but on the perdurantist view, the temporal parts are not the ones that are moving
 - On the view of carving up spatio-temporal subregions of the disk which corresponds to a part of the disc: the two discs have duplicate spatial parts and each disc has parts that behave in a way we would expect a stationary's disc's part to behave, as well as ones that behave the way we expect a rotating disc's to behave
- **Perdurantist Revision:**
 - Actually difficult for Perdurantists to achieve this result
 - To see why: The liberalist endurantist faces the same problems because they accept the existence of the following objects
 - SS1 (small stationary 1): occupies p1 at t1 and p1 at t2
 - SS2 (small stationary 2): occupies p2 at t1 and p2 at t2
 - SM1 (small moving 1): occupies p1 at t1 and p2 at t2
 - SM2 (small moving 2): occupies p2 at t1 and p1 at t2
 - Large: Occupies both points at both times
 - Leaves open which of the smaller objects if any are part of Large (and so is a coherent endurantist view) → even without positing any restrictions on which objects the laws of motion apply to, we can describe two distinct possibilities: one involving the object Large having SS1 and SS2 as parts (Large is stationary), and one in which Large has SM1 and SM2 as parts (Large is rotating)
 - Difficult for liberal perdurantist to achieve this result: for if the disc is stationary, then SS1 and SS2 are parts of it both at t1 and t2 → but by the perdurantist definition of part-at-a-time, this requires that SS1's ITP at t1, which requires that SS1's ITP at t1 (an object which exists only at t1 and

occupies only p1) be part of the disc's ITP at t1; similarly SS2's ITP at t2 must be part of the disc's ITP at t2

- Considering SM1 which shares SS1's ITP at t1 and it shares SS2's ITP at t2

- So SM1 must be part of the station disc at both times
- Restrictive endurantism can just deny the existence of an object which occupies the spatio-temporal region of O1 at t and the distant O2 at time shortly after t → can construe the two discs as having crucially different profiles where the spacetime region occupying the top-left quadrant of the disc at t1 does not change in t2 (for the stationary disc) and changes at t2 for the moving disc

7. Argument from Permanent Coincidence (Pro-Endurantism)

- **Objection (Endurantist advantage):**
 - Endurantism better accommodates permanent coincidence (e.g., two permanently coinciding objects) without invoking temporal parts.
- **Perdurantist Revision:**
 - Perdurantists can revise their theory by allowing permanently coincident objects to share all temporal parts while differing via modal or relational properties, thus preserving distinct identities without contradiction.